A second antipodal 840-point kissing configuration in $\mathbb{R}^{12}$, slice-closure theorems, and the architecture frontier at 842

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Abstract

We exhibit a new antipodal kissing configuration of 840 points (420 lines through the origin of $\mathbb{R}^{12}$ at pairwise angles at least $60°$): the 132 root lines $(e_a \pm e_b)/\sqrt{2}$ together with 288 half-integer lines with entries $\pm\frac{1}{2\sqrt{2}}$ on eight-coordinate supports, in the even sign class, spread over 15 partial cells. All $\binom{420}{2} = 87{,}990$ pairwise inner products are certified $|\langle u, v\rangle| \le 1/2$ in exact arithmetic by two independent routes, of which the stdlib checker ships with this note's repository; 20,464 pairs (23.3%) meet the bound with equality, so the configuration is extremely rigid. A packing-independent counting theorem proves it is *not* isometric to the classical 840-point configuration (24 axis vectors plus 816 weight-4 vectors over a maximum quadruple packing), nor to any configuration of 12 axes plus 51 whole weight-4 cells: the new inner-product histogram $\{0 : 35{,}142,\ \frac{1}{4} : 32{,}384,\ \frac{1}{2} : 20{,}464\}$ forces a block-pair statistic $P_2 \ge 564$ while matching it requires $P_2 = 563$. An exact $\mathbb{Q}(\sqrt{2})$ rotation certificate nevertheless shows the new configuration is a blockwise $45°$ rotation of a 12-axes + 47-whole + 8-half weight-4-cell configuration: it evades the known 420 but not the integer weight-4 world, and is (to our knowledge) the first 420-line configuration that uses *partial* cells. We prove a slice-closure theorem, whose single computational input is a CP-SAT OPTIMAL certificate: for every one of the 10,395 pairings of the 12 coordinates into 6 pairs, the root lines together with the pair-aligned half-integer cells embed isometrically into the integer world, so such "slice" configurations cap at exactly 420 lines and pair-aligned half-integer configurations cap at exactly 288. An exact CP-SAT ladder then proves (OPTIMAL at every node, full orbit enumeration through level 3) that any 289-line half-integer configuration — which would yield 421 lines, 842 points, and a world-record kissing configuration — must use at least 4 cells foreign to *every* pairing; extensive exact search at levels 4–6 and beyond found none. We state the honest scope precisely: α(half-integer world) = 288 is strongly evidenced but not globally proven. A structural attack on the remaining gap then yields four theorems: an exact structure lemma for conflicting cell pairs (the bipartite conflict graph at support overlaps 5/6/7 is $16 \times K_{4,4}$, $32 \times K_{2,2}$, and the folded-7-cube unit-ball bipartite graph); a closure of the even-overlap side proving (given the same solver-certified slice cap) that the missing *odd-overlap*

*Correspondence: `kevinrussell@gmail.com`. Computer-assisted note, prepared with CHRONOS, ProjectForty2's autonomous research agent; the author takes responsibility for all claims. The configuration was found by large-scale independent-set search (KaMIS [11]) over an exactly generated conflict graph; every *certified* claim in this note reduces to integer or $\mathbb{Q}(\sqrt{2})$ arithmetic that the accompanying stdlib-only checker recomputes from the raw witness (Appendix A), and every *solver-status* claim (CP-SAT OPTIMAL, certified Lovász ϑ) is identified as such in the text and shipped with its receipt. This is the third note of an exact-certificate series [16, 17].

lemma is *equivalent* to $\alpha(U) = 288$; an LP barrier theorem — no family of valid occupancy inequalities with at most 109 positive-support cells each (which includes every pair, triple, localization, slice, and ladder cut we possess) can certify $\alpha(U) \le 289$; and a star reduction $\alpha(U_{12}) \le 3\,\alpha(U_{11})$ that passes the barrier and is exactly tight on the witness, reducing both the global closure and the record hunt to a single question in a world three times smaller: is $\alpha(U_{11}) \le 96$? Finally, a ϑ-certified architecture sweep closes every enumerable frame family in dimensions 11 and 12 far below the record thresholds and exposes a structural anti-correlation: the richer the frame, the thinner the weight-4 base it leaves behind. Whether $\mathbb{R}^{12}$ contains an antipodal 842-point kissing configuration (421 lines; our prior LP ceiling is 614 lines [17]) remains open.

1 Introduction

The kissing number $k(d)$ is the maximum number of unit spheres that can simultaneously touch a central unit sphere in $\mathbb{R}^d$; equivalently, the maximum number of unit vectors with pairwise angles at least 60°. In $\mathbb{R}^{12}$ the classical record configuration has 840 points ([6, Table 1.5]; the kissing-number survey [3] confirms that the $n \le 24$ lower-bound table, including 840 at $n = 12$, follows SPLAG Table 1.5), and the record was recently raised to $k(12) \ge 841$ by Takhanov, Assylbekov and Yun [18], who found room for an 841st sphere by a global rebuild of an 840-point base; the best upper bound is $k(12) \le 1355$ [12]. Neighboring dimension 11 has seen the same burst of computer-search activity ($k(11) \ge 593$ by AlphaEvolve [1], then $k(11) \ge 604$ by Bianchi et al. [2]).

A kissing configuration is *antipodal* if it is closed under $x \mapsto -x$; antipodal configurations of $2N$ points are the same thing as systems of N lines through the origin with pairwise angles at least 60° (coherence $|\cos| \le 1/2$). The 840-point classical record configuration is antipodal (420 lines); the 841-point record of [18] is *not*: an odd-cardinality set cannot be closed under $x \mapsto -x$, and our direct check of coordinates recovered from the authors' published Gram data — a floating-point side observation, not part of this note's certificate set — finds no exactly antipodal pair (minimum pairwise cosine -0.99999999; the global rebuild retains hundreds of *near*-antipodal pairs but exact antipodality appears to be broken). An antipodal configuration with 842 points is exactly a system of 421 lines at coherence 1/2; since $842 > 841$, *any* such system would be a world-record kissing configuration. In our predecessor note [17] we proved the two-point LP ceiling $N_{60}(12) \le 614$ in exact rational arithmetic — so an antipodal record attempt at 421 lines is not excluded by any known ceiling; the gap between the realized 420 and the LP 614 is enormous. The present note documents what we found when we went looking for the 421st line.

The contributions are:

1. **A new 420-line configuration** (Section 3), found by independent-set search over an exactly generated mixed-alphabet conflict graph and verified twice in exact arithmetic (Theorem 1). Its composition — 132 root lines plus 288 half-integer lines over 15 *partial* cells, zero axes, zero weight-4 lines — is structurally unlike the classical 420.

2. **A novelty theorem** (Section 4): a packing-independent counting argument (Theorem 2) proving non-isometry to the classical configuration and to every configuration of its combinatorial type. The proof is a five-line double count; no solver is involved.

3. **A rotation certificate and slice-closure theorems** (Section 5): an exact $\mathbb{Q}(\sqrt{2})$ isometry mapping the new configuration into the integer weight-4 world (Theorem 5), and a closure theorem for every coordinate pairing (Theorem 6) showing that "slice" configura-

tions cap at exactly 420 lines. The new configuration therefore ties the record *within* a capped world; any improvement must mix across slices.

4. **An exact exclusion ladder** (Section 6): CP-SAT proofs, OPTIMAL at every node with full orbit enumeration through level 3, that a 289-line half-integer configuration must sit at least 4 cells foreign to every one of the 10,395 pairings (Theorem 8), together with the (negative) outcome of extensive exact search beyond the proven levels. We state exactly what is proven and what is only searched.

5. **The odd-overlap frontier** (Section 7): an exact structure lemma for every conflicting cell pair (Theorem 9) with machine-tight pair and triple caps; a proof that configurations whose occupied cells overlap pairwise *evenly* cap at 288 (Theorem 17), repairing a false cell-level claim in our own campaign roadmap; the resulting equivalence *odd-overlap lemma* $\Leftrightarrow$ $\alpha(U) = 288$ (Corollary 19); an LP barrier theorem (Theorem 22) proving that no aggregation of small-support valid inequalities — including every cut family in this note — can certify $\alpha(U) \leq 289$; and a star reduction (Theorem 24) to the 10,560-line U_{11} world that passes the barrier and is exactly tight on the witness. The paper's headline open problem becomes: *is* $\alpha(U_{11}) \leq 96$*?*

6. **An architecture-exhaustion sweep** (Section 8): ϑ-certified and ILP-exact closures of every enumerable frame family we could write down in dimensions 11 and 12, all far below the record thresholds, and the structural anti-correlation between frame richness and weight-4 base capacity that explains the pattern.

Section 9 collects open problems, chief among them the U_{11} question $\alpha(U_{11}) \leq 96$ — to which Section 7 reduces both the global closure of the half-integer world *and* the antipodal record hunt — and the status of 842.

Throughout, "certified" means: recomputed from the raw witness by the stdlib-only checker in exact integer or $\mathbb{Q}(\sqrt{2})$ arithmetic (`make verify`, Appendix A); no floating point on the certified path, and no claimed value echoed from a file. Solver statuses (CP-SAT OPTIMAL, certified ϑ, KaMIS incumbents) are reported as such.

2 Four alphabets and one conflict graph

All configurations in this note live in the union of four line alphabets in $\mathbb{R}^{12}$, each with an integer representative:

alphabet	unit representative	integer rep. g	$N = \langle g, g \rangle$	#lines
A (axes)	e_i	e_i	1	12
W_2 (roots)	$(e_a \pm e_b)/\sqrt{2}$	$e_a \pm e_b$	2	132
W_4 (weight-4)	$\frac{1}{2}\sum_{i\in B} s_i e_i$, $\lvert B\rvert = 4$, $s_i = \pm 1$	$\sum_{i\in B} s_i e_i$	4	3960
U (half-integer)	$\frac{1}{2\sqrt{2}}\sum_{i\in H} s_i e_i$, $\lvert H\rvert = 8$, even	$\sum_{i\in H} s_i e_i$	8	31,680

Here B ranges over the $\binom{12}{4} = 495$ four-point supports (8 lines each, sign vectors up to negation), and H over the $\binom{12}{8} = 495$ eight-point supports ("cells"). In the U alphabet we keep only the *even sign class* (an even number of minus signs); negation preserves sign parity on an 8-support, so each cell carries $2^8/2 \cdot \frac{1}{2} = 64$ lines and $|U| = 495 \cdot 64 = 31{,}680$.

Two lines with integer representatives g_1, g_2 of norms N_1, N_2 are *in conflict* (angle below 60°) iff

$$4\,\langle g_1, g_2\rangle^2 \;>\; N_1 N_2, \tag{1}$$

a pure big-integer predicate; a legal configuration is an independent set of the conflict graph. The full graph on 12+3960+132+31,680 = 35,784 vertices has 40,130,904 edges. Some structural facts, each recomputed exactly from (1) during graph construction (artifact `K2_stats.json`, predicate-validation mismatches 0; the repository README reproduces the full alphabet-pair edge census):

- W_2 is internally conflict-free (0 W_2–W_2 edges) and conflict-free against all of U (0 W_2–U edges); each root line conflicts with exactly the two axes in its support (264 A–W_2 edges), and with weight-4 lines as dictated by (1) (23,760 W_4–W_2 edges).

- Two U-lines in cells H, H' with $s = |H \cap H'|$ common coordinates have $\langle u, u'\rangle = \frac{1}{8}\sum_{i\in H\cap H'} s_i s'_i$, so they conflict iff $\left|\sum_{i\in H\cap H'} s_i s'_i\right| \geq 5$; all realized $|\cos|$ values lie in $\{0, \frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{1}{2}\}$. Cell pairs with $s = 4$ are conflict-free; the census of cell pairs is s=4: 17,325, s=5: 55,440, s=6: 41,580, s=7: 7,920 (total $\binom{495}{2}$ = 122,265). A conflicting cell pair induces 256/128/512 conflict edges among its 64×64 line pairs at $s = 5/6/7$ respectively (the structure lemma, Theorem 9), and at most 64 lines can be chosen from the two cells jointly (the pair cap, Corollary 10).

- The induced U-world graph has 31,680 vertices, 23,569,920 edges, and is 1488-regular and vertex-transitive; its largest adjacency eigenvalue is 1488 (the regularity, exact), and its smallest is -50 by numerical eigensolve (`V2_DECISION_SUMMARY.json`; we have not certified $\lambda_{\min}$ exactly, and it enters only the avowedly loose Hoffman row of Table 2).

Because coherence is scale-invariant, mixing the norm-2, norm-4 and norm-8 shells is legitimate — the mixed graph is exactly the object a cross-shell ("grand mixed") search should explore, and to our knowledge it had not been searched before.

3 The configuration and its verification

Theorem 1 (a second 420)**.** *There exists an antipodal kissing configuration of* 840 *points in* $\mathbb{R}^{12}$*, i.e. a system of* 420 *lines with pairwise* $|\cos| \leq 1/2$*, consisting of all* 132 *root lines* W_2 *together with* 288 *half-integer lines from* U*, distributed over the* 15 *cells of Table 1 (*24 *lines in each of* 3 *cells,* 20 *in each of* 6*,* 16 *in each of* 6*). All* 87,990 *line pairs satisfy the coherence bound exactly, with* 20,464 *pairs at* $|\cos| = \frac{1}{2}$ *exactly, and the multiset of* $|\cos|$ *values is*

$$\left\{0^{\,35142},\ \left(\tfrac{1}{4}\right)^{32384},\ \left(\tfrac{1}{2}\right)^{20464}\right\}.$$

Proof. `make verify`. The witness `certs/config420.json` lists the 420 integer representatives; the checker recomputes every pairwise predicate (1) in big-integer arithmetic (independently re-verified in exact $\mathbb{Q}(\sqrt{2})$ by the campaign adjudicator, a second route that is not part of this repository's certificate set: 0 violations on both), together with the composition, the cell distribution, and the displayed histogram. No claimed count is echoed: histogram and margins are recomputed from the vectors. □

Provenance (not part of the certificate). The configuration was found by KaMIS [11] (weighted branch-and-reduce, 7200 s) as a maximum-independent-set incumbent on the full 35,784-vertex mixed graph. The incumbent contains zero axes and zero weight-4 lines — the search abandoned

cell support H	lines	cell support H	lines
$\{0,1,2,5,6,7,9,11\}$	24	$\{1,2,3,4,6,7,8,10\}$	20
$\{0,3,4,5,8,9,10,11\}$	24	$\{0,1,2,3,4,5,9,11\}$	16
$\{1,2,3,4,5,6,7,11\}$	24	$\{0,1,2,3,4,6,7,9\}$	16
$\{0,1,2,3,4,8,9,10\}$	20	$\{0,1,2,5,8,9,10,11\}$	16
$\{0,1,2,6,7,8,9,10\}$	20	$\{0,3,4,5,6,7,9,11\}$	16
$\{0,3,4,6,7,8,9,10\}$	20	$\{1,2,5,6,7,8,10,11\}$	16
$\{0,5,6,7,8,9,10,11\}$	20	$\{3,4,5,6,7,8,10,11\}$	16
$\{1,2,3,4,5,8,10,11\}$	20		

Table 1: The 15 cells of the new configuration and their line counts ($3\times 24+6\times 20+6\times 16 = 288$). Every cell is *partial* (out of 64 possible lines), and — as Section 5 explains — the 15 supports are exactly the $\binom{6}{4} = 15$ supports aligned with one pairing of the 12 coordinates into 6 pairs.

the classical architecture entirely. Two facts make the object remarkable before any novelty argument: 23.3% of its pairs are at zero margin (tight $|\cos| = \frac{1}{2}$), and its 288 half-integer lines occupy 4.5 cells' worth of capacity spread over 15 partial cells, so within the half-integer world *mixing beats whole-cell packing* — the opposite of the regime we had established in dimension 11.

Since the 132 root lines are conflict-free among themselves and against all of U, this architecture reduces cleanly:

$$\alpha\big(W_2 \cup U\big) \;=\; 132 + \alpha(U), \tag{2}$$

and the record condition "421 lines" is exactly $\alpha(U) \geq 289$. The rest of the note is organized around that single question.

4 Novelty: a packing-independent counting theorem

The classical 420 consists of the 12 axes plus 51 *whole* weight-4 cells over a maximum quadruple packing: $D(3,4,12) = 51$ is the maximum number of 4-subsets of a 12-set with every 3-subset covered at most once. In constant-weight-code terms this is $A(12,4,4) = 51$, listed as exact in the standard tables [4]; quadruple systems go back to Hanani [8], and [9] surveys the packing theory. Our artifacts independently cross-check the value by exact ILP (HiGHS, integer optimal). With 51 blocks, $12 + 8\cdot 51 = 420$ lines and $24 + 16\cdot 51 = 840$ points. The inner-product histogram is an isometry invariant, and the classical configuration's histogram, recomputed exactly from an explicit 51-block packing (artifact `modular420_invariants.json`; a campaign artifact, not part of this repository's certificate set), is $\{0 : 35{,}238,\ \frac{1}{4} : 32{,}256,\ \frac{1}{2} : 20{,}496\}$ — already different from the new configuration's histogram, which settles non-isometry to that specific object. The following theorem is much stronger: no configuration of the classical *type* realizes the new histogram, whatever packing it is built on.

Theorem 2 (novelty)**.** *No line configuration consisting of the* 12 *axis lines together with* 51 *whole weight-*4 *cells (over any legal family of* 51 *blocks) has inner-product histogram* $\{0 : 35{,}142,\ \frac{1}{4} : 32{,}384,\ \frac{1}{2} : 20{,}464\}$*. Consequently the configuration of Theorem 1 is not isometric to the classical* 420*, nor to any axes-plus-whole-cells configuration.*

Proof. Every line of such a configuration has an integer representative of norm $N \in \{1,4\}$, so every pairwise $|\cos|$ is $|d|/\sqrt{N_1 N_2}$ with $d \in \mathbb{Z}$ and $\sqrt{N_1 N_2} \in \{1,2,4\}$: a multiple of $\frac{1}{4}$, and legality restricts to $\{0, \frac{1}{4}, \frac{1}{2}\}$.

Let the blocks be $B_1, \dots, B_{51}$, let r_i be the number of blocks containing point i, and let P_1 (resp. P_2) be the number of block pairs sharing exactly 1 (resp. 2) points. Legality forces every block pair to share at most 2 points: two *whole* cells on blocks sharing 3 points contain a sign-aligned triple, realizing $|\cos| = \frac{3}{4}$. Counting each $|\cos|$ class (a finite check per case; the checker recomputes it): axis–axis pairs contribute 0's; axis–cell pairs contribute $|\cos| = \frac{1}{2}$ exactly when the axis point lies in the block (8 lines per block): $8\sum_i r_i$ pairs; each whole cell internally contributes 16 pairs at $\frac{1}{2}$ and 12 at 0; a block pair sharing 2 points contributes 32 pairs at $\frac{1}{2}$ and 32 at 0; a block pair sharing 1 point contributes all 64 pairs at $\frac{1}{4}$; disjoint block pairs contribute 0's. Hence

$$n_{1/4} = 64\,P_1, \qquad n_{1/2} = 8\sum_i r_i + 16\cdot 51 + 32\,P_2.$$

Since every block has 4 points, $\sum_i r_i = 4\cdot 51 = 204$ always, so matching the stated histogram requires

$$P_1 = \tfrac{32384}{64} = 506, \qquad P_2 = \tfrac{20464-1632-816}{32} = 563.$$

Double-counting incidences of points in block pairs, $\sum_i \binom{r_i}{2} = P_1 + 2P_2 = 506 + 1126 = 1632$. By convexity with $\sum_i r_i = 204$ over 12 points (mean 17), $\sum_i \binom{r_i}{2} \geq 12\binom{17}{2} = 1632$, with equality iff $r_i = 17$ for every i. So $r_i \equiv 17$ is forced.

Now let λ_p be the number of blocks containing the pair $p \subset [12]$. Blocks through a fixed pair p have pairwise-disjoint 2-point remainders in the other 10 points (two blocks through p sharing a third point would share 3 points), so $\lambda_p \leq 5$. Fix a point i: each block through i contains 3 pairs through i, so $\sum_{j\neq i}\lambda_{ij} = 3r_i = 51$, a sum of 11 values in $\{0,\dots,5\}$. The convex functional $\sum_{j\neq i}\binom{\lambda_{ij}}{2}$ is minimized at the most-equal admissible vector $(5^7, 4^4)$: $7\cdot 10 + 4\cdot 6 = 94$. A block pair sharing exactly 2 points shares exactly one pair, so $\sum_p \binom{\lambda_p}{2} = P_2$, and summing the per-point inequality over the 12 points counts each pair twice:

$$2P_2 = \sum_i \sum_{j\neq i} \binom{\lambda_{ij}}{2} \;\geq\; 12\cdot 94 = 1128, \qquad \text{so } P_2 \geq 564.$$

This contradicts $P_2 = 563$. □

Remark 3. The bound is tight in the right way: the classical configuration itself has $(P_1, P_2) = (504, 564)$ with $r_i \equiv 17$ and pair degrees $\lambda \in \{4^{24}, 5^{42}\}$, sitting exactly on the convexity floor. The new histogram misses it by one.

Remark 4 (weaker invariants fail). The novelty is invisible to cruder tests. The new configuration realizes only the classical $|\cos|$ support $\{0, \frac{1}{4}, \frac{1}{2}\}$ — the odd-eighth values $\frac{1}{8}, \frac{3}{8}$ available to the half-integer alphabet are never used (every cohabiting cell pair has even overlap), so the "odd-eighth certificate" one might hope for does not fire. Per-line degree profiles also fail: the new configuration contains exactly 12 lines with profile $(n_0, n_{1/4}, n_{1/2}) = (283, 0, 136)$, *identical* to the classical configuration's axis lines at replication 17. The counting theorem is what remains after these invariants collapse.

5 The rotation certificate and slice closure

Novelty notwithstanding, the new configuration is not far from the integer world — it is a rotation of a mixed whole/partial-cell integer configuration, a fact we certify exactly.

For a *pairing* P of the 12 coordinates into 6 unordered pairs (there are $11!! = 10{,}395$ pairings), let $T_P \in O(12)$ be the blockwise rotation by 45° in each of the 6 coordinate planes: on the plane of the pair (a, b),

$$T_P\, e_a = \tfrac{1}{\sqrt{2}}(e_a + e_b), \qquad T_P\, e_b = \tfrac{1}{\sqrt{2}}(-e_a + e_b).$$

Call an 8-support H *P-aligned* if H is a union of 4 pairs of P (equivalently its 4-point complement is a union of 2 pairs); each pairing has $\binom{6}{4} = 15$ aligned supports.

Theorem 5 (rotation certificate)**.** *Let $P_0 = \{(0, 9), (1, 2), (3, 4), (5, 11), (6, 7), (8, 10)\}$. Then T_{P_0} maps the configuration of Theorem 1 exactly — certified in $\mathbb{Q}(\sqrt{2})$ arithmetic, line by line — onto a configuration in the integer world $A \cup W_4$ consisting of the 12 axis lines plus 408 weight-4 lines over 55 supports: 47 whole cells (8 lines) and 8 half cells (4 lines).*

Proof. `make verify` replays the isometry: the checker applies the explicit T_{P_0} (entries in $\frac{1}{\sqrt{2}}\mathbb{Z}$) to each of the 420 integer representatives, working in $\mathbb{Z}[\sqrt{2}]$, and verifies that every image is an axis or weight-4 line, then recomputes the support census $47 \cdot 8 + 8 \cdot 4 = 408$, $12 + 408 = 420$ (certificate `certs/rotation.json`). □

So the new configuration is Hanani-capped in disguise: up to isometry it lives inside the integer weight-4 world — but as the first known 420 using *partial* cells (8 half cells), which is exactly what Theorem 2 says it must do to avoid the classical type. The mechanism generalizes to every pairing:

Theorem 6 (slice closure)**.** *Fix any pairing P and let the P-slice be the line family consisting of all 132 root lines together with all $15 \cdot 64 = 960$ half-integer lines in P-aligned cells. Then:*

(a) T_P maps the P-slice isometrically into the integer world $A \cup W_4$;

(b) every legal configuration of P-aligned half-integer lines has at most 288 lines, and 288 is achieved;

(c) every legal configuration inside the P-slice has at most 420 lines, and 420 is achieved.

Proof. (a) Root lines on a pair of P map to axes: $T_P\big((e_a{+}e_b)/\sqrt{2}\big) = e_b$ and $T_P\big((e_a{-}e_b)/\sqrt{2}\big) = e_a$ for $(a, b) \in P$ (12 lines); root lines across two pairs map to weight-4 lines (four entries $\pm\frac{1}{2}$). For a half-integer line u on a P-aligned support, each of its 4 pairing planes carries coordinates $(x, y) \in \{\pm\frac{1}{2\sqrt{2}}\}^2$, and in-plane

$$T_P:\ (x, y)\ \longmapsto\ \Big(\tfrac{x-y}{\sqrt{2}},\ \tfrac{x+y}{\sqrt{2}}\Big),$$

where exactly one of $x \pm y$ vanishes and the other is $\pm\frac{1}{\sqrt{2}}$; so the image has exactly one entry $\pm\frac{1}{2}$ per plane: four entries $\pm\frac{1}{2}$ on a 4-support, a weight-4 line. An orthogonal map preserves all inner products, and distinct lines map to distinct lines.

(b) Upper bound: the P-aligned half-integer world is a 960-vertex induced subgraph of the U-conflict graph; an exact CP-SAT [14] solve certifies its maximum independent set to be 288 (status OPTIMAL; receipt `certs/receipts/result_slice_only.json`). The symmetric group S_{12} permutes coordinates, acts transitively on pairings, and induces isomorphisms of the aligned-cell graphs, so the computation for one pairing settles all 10,395. Achievability is the 288-line witness of Theorem 1 (whose 15 cells are exactly the P_0-aligned supports — all of them carry lines).

(c) Root lines are conflict-free among themselves and against all of U, so a maximum slice configuration is all 132 roots plus a maximum P-aligned U-configuration: $132 + 288 = 420$ by (b), achieved by Theorem 1. □

Remark 7 (two routes to the cap). Part (b) rests on one finite certified-OPTIMAL computation. An alternative, computation-free route runs through (a) plus the maximality of the classical 840 for the integer norm-4 alphabet ($\alpha(A \cup W_4) = 12 + 8 \cdot D(3, 4, 12) = 420$): embedding a P-aligned 289-configuration plus all 132 roots into $A \cup W_4$ would give 421 there. We established that maximality in earlier campaign work (campaign log, 2026-07-01); its formal writeup and receipt are *not* part of this note's certificate set, which is why the theorem above is stated with the self-contained CP-SAT route. (A subtlety worth recording: *partial* W_4 cells can share 3 support points without illegality — only two *whole* cells on 3-point-sharing blocks force a sign-aligned triple — so the $A \cup W_4$ cap is genuinely stronger than the naive whole-cell count and must be proven at line level, as the campaign computation did.)

The strategic consequence is sharp: within any single slice the record is unreachable (420 exactly), so

a 421*-line configuration of this architecture — equivalently a* 289*-line half-integer configuration, by* (2) *— must mix across slices, i.e. use cell pairs with odd overlap* $|H \cap H'| \in \{5, 7\}$*, the* $|\cos| \in \{\frac{1}{8}, \frac{3}{8}\}$ *regime.*

The "i.e." deserves care: escaping every slice and containing an odd-overlap cell pair are *not* the same condition at cell level, and the gap is exactly the even-but-unaligned cell families of Section 7, where the displayed statement is proven outright (Corollary 18).

6 The ladder: what is proven about $\alpha(U)$, and what is only searched

Fix the pairing P_0 and call a cell *foreign* if it is not P_0-aligned ($495 - 15 = 480$ foreign cells). Let $B_6 \leq S_{12}$ be the stabilizer of P_0 (the hyperoctahedral group of order $2^6\, 6! = 46{,}080$), which acts on foreign cells and their tuples. The exact ladder solves, with CP-SAT to certified OPTIMAL, the maximum independent set of the P_0-slice augmented by k foreign cells, for *every* orbit of k-tuples at levels $k \leq 3$:

level	universes solved	orbit enumeration	result	status
0	slice only (960 v.)	—	288	OPTIMAL
1	all 480 foreign cells	exhaustive	288 in all 480	all OPTIMAL
2	39 B_6-orbit reps of foreign pairs	complete	288 in all 39	all OPTIMAL
3	1284 B_6-orbit reps of foreign triples	complete	288 in all 1284	all OPTIMAL

Theorem 8 (exclusion ladder). *Every legal half-integer configuration all of whose cells lie within* 3 *foreign cells of some pairing — i.e. every configuration whose cell set is contained in (aligned cells of P) $\cup$ ($\leq$ 3 other cells), for some pairing P — has at most* 288 *lines. Equivalently: a* 289*-line half-integer configuration must use at least* 4 *cells foreign to every one of the* 10,395 *pairings.*

Proof. By S_{12}-transitivity on pairings it suffices to treat P_0. A configuration using at most 3 foreign cells is contained in one of the level-≤ 3 universes, each of whose maximum independent sets is certified OPTIMAL at 288; the B_6-orbit enumeration at levels 2 and 3 is complete (level-3 orbits computed by connected components under a 3-element generating set of B_6 and validated by reproducing the 39 level-2 orbits with the same machinery). Receipts, all under `certs/receipts/`:

`result_slice_only.json`, `level2.json` and `level3.json` for levels 0/2/3; the level-1 per-cell results (all 480 foreign cells at 288, all OPTIMAL) are recorded in `V2_DECISION_SUMMARY.json` (`exact_results.LEVEL1_complete`). □

Beyond level 3 *(search, not proof).* Random level-4/5/6 universes (≈ 120 each: 119/118/117 certified OPTIMAL at 288, the remainder timing out with incumbent 288, zero records); all 31 two-slice unions $P_0 \cup P_1$ with P_1 one element-swap away (best 288, 24 of 31 OPTIMAL); anchored 100-cell universes around the slice (20+ diverse runs, never above 288); free random 100-cell universes (only ≈ 142 — structure-poor, confirming that slices are where the capacity lives); and long multi-seed KaMIS runs on the full U-world, which reproduce 288 and never 289. The 288 witness was accepted as a warm start in every bounded run and never improved.

Why no global theorem yet. The full U-world exact model (31,680 binary variables, 23,569,920 constraints) exceeded the memory budget of our shared hardware; CP-SAT's dual bound degrades quickly beyond ≈ 27-cell universes; and every tractable convex relaxation has a hopeless integrality gap (Table 2). The honest summary:

> $\alpha(U) \geq 288$ is certified (explicit witness). $\alpha(U) = 288$ is **proven for every configuration within** 3 **foreign cells of any slice** (Theorem 8) and **unrefuted by extensive exact search** beyond that radius, but it is **not globally proven**. Every "289-witness or world record" statement in our campaign remains conditional on closing this gap.

relaxation	bound on $\alpha(U)$	nature
Delsarte angle LP [7] ($\lvert\cos\rvert \in \{0, \frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{1}{2}\}$)	≤ 583.7	numeric, converged
Hoffman ratio $n(-\lambda_{\min})/(d - \lambda_{\min})$, $d = 1488$, $\lambda_{\min} = -50$	≤ 1029.91	spectral
LP with all 10,395 slice cuts (≤ 288 per pairing)	9504	fractional ($x \equiv 0.3$)

Table 2: Rigorous-but-loose global upper bounds versus the believed truth 288: the integrality gap is enormous, so no tractable convex relaxation can decide 289. Decimals rounded toward $+\infty$. The Delsarte value is the converged floating-point LP optimum (583.68), *not* an exact-rational certificate like those of the predecessor note [17], and the Hoffman row uses the numerically computed $\lambda_{\min} = -50$ of Section 2; both rows are loose by a factor ≥ 2, so their numeric status is immaterial to every conclusion. Theorem 22 explains the third row's uselessness structurally.

7 The odd-overlap frontier: pair structure, even closure, an LP barrier, and the star reduction

The ladder localizes a hypothetical 289; this section reports a direct structural attack on the gap itself (a dedicated lemma-hunt session, 2026-07-10/11). The attack did not prove the odd-overlap lemma. It produced something arguably better: an exact description of the local conflict structure, a closure of the entire even-overlap side that repairs a hole in our own roadmap, a proof that the odd-overlap lemma is *equivalent* to the global statement $\alpha(U) = 288$, a barrier theorem showing that every cut family in this note — the ladder's included — is provably insufficient to close it, and a reduction of both the closure and the record hunt to a single sharp question in a world three times smaller.

Every claim below carries one of three labels. (PROVEN): a complete proof appears here. (MACHINE): an exhaustive exact computation — stdlib-only big-integer branch-and-bound, no solver, no floats — cited with its exact content; the scripts and result files ship with the repository (`scripts/oddlemma/`) and every such entry reproduces in seconds. (OPEN): open, stated as such.

Notation. We index a cell by its 4-point *defect* $A = [12] \setminus H$ rather than its 8-point support H; cells are the $\binom{12}{4} = 495$ four-subsets of [12]. For cells $A \neq B$ put $t = |A \cap B|$; the common support is exactly $H \cap H' = [12] \setminus (A \cup B) =: R$ with $r = |R| = 4 + t$, so $t = 0, 1, 2, 3$ corresponds to support overlap $|H \cap H'| = 4, 5, 6, 7$. $t = 0$ pairs are conflict-free ("free"), $t \in \{1, 3\}$ are the odd-overlap pairs, $t = 2$ the even ones. For a configuration S and a cell A, n_A denotes the number of S-lines in cell A ($0 \leq n_A \leq 64$).

7.1 The structure of a conflicting cell pair

Theorem 9 (structure lemma)**.** *Let A, B be cells with $t = |A \cap B| \in \{1, 2, 3\}$, and for a line u in either cell let its color be the restriction of its sign vector to R, taken modulo global sign. Then the bipartite conflict graph between the 64 lines of A and the 64 lines of B is:* (i) *for $t = 1$ ($r = 5$): 16 color classes of 4 lines per cell; two lines conflict iff they have the same color; the graph is 16 disjoint copies of $K_{4,4}$, 256 edges.* (ii) *for $t = 2$ ($r = 6$): 32 colors of 2; 32 disjoint $K_{2,2}$, 128 edges.* (iii) *for $t = 3$ ($r = 7$): the color map is a bijection from each cell's 64 lines onto the folded 7-cube $\mathbb{F}_2^7/\langle \mathbf{1} \rangle$, and two lines conflict iff their folded Hamming distance is ≤ 1; the graph is the bipartite unit-ball graph of the folded 7-cube, $64 \cdot 8 = 512$ edges. (PROVEN; censuses 256/128/512 independently MACHINE-verified, `scripts/oddlemma/uw.py` self-test, and recorded in the campaign artifact `V2_DECISION_SUMMARY.json`)*

Proof. Write $\varepsilon, \varepsilon'$ for sign vectors of lines u (on H_A) and v (on H_B). Since $8\langle u, v \rangle = \sum_{i \in R} \varepsilon_i \varepsilon'_i$ and conflict means $|\cos| > \frac{1}{2}$, the pair conflicts iff $\left|\sum_{i \in R} \varepsilon_i \varepsilon'_i\right| \geq 5$. With d the Hamming distance of the two restrictions to R, the sum is $r - 2d$, so conflict means $|r - 2d| \geq 5$: for $r = 5$, $d \in \{0, 5\}$; for $r = 6$, $d \in \{0, 6\}$; for $r = 7$, $d \in \{0, 1, 6, 7\}$. The first two cases say exactly "restrictions agree up to global sign on R" (same color); the third says "folded distance ≤ 1".

Fiber sizes: a cell's support H contains R with $|H \setminus R| = 8 - r$. Given any sign pattern on R, the even-parity constraint leaves 2^{8-r-1} completions on $H \setminus R$ when $r < 8$, and modding by global flip pairs the pattern x (with its completions) against $\bar{x}$ (with the flipped completions); each folded color class $\{x, \bar{x}\}$ therefore carries exactly 2^{8-r-1} lines of the cell: 4 per class for $r = 5$ (16 classes), 2 for $r = 6$ (32 classes), 1 for $r = 7$ (64 classes — a bijection). The folded 7-cube is 7-regular (neighbors of $\{x, \bar{x}\}$ are the classes $\{x + e_i\}$, $i \in R$, pairwise distinct), so each line has $1 + 7 = 8$ conflict partners at $t = 3$, giving $64 \cdot 8 = 512$ edges; the same-color counts give $16 \cdot 4 \cdot 4 = 256$ and $32 \cdot 2 \cdot 2 = 128$. □

Corollary 10 (pair cap)**.** *For every conflicting cell pair, $n_A + n_B \leq 64$, and 64 is attained for every $t \in \{1, 2, 3\}$. (PROVEN)*

Proof. $t = 1, 2$: within one color class the conflict graph is complete bipartite, so an independent set uses lines of at most one cell per class; each class holds 4 (resp. 2) lines of one cell, whence at most $16 \cdot 4$ (resp. $32 \cdot 2$) $= 64$ lines in total. $t = 3$: identify both cells with the folded 7-cube V; an independent set is $S_A \sqcup S_B$ with $S_B \cap N[S_A] = \emptyset$, so $|S_A| + |S_B| \leq |S_A| + 64 - |N[S_A]| \leq 64$ since $S_A \subseteq N[S_A]$. Tightness: one full cell. □

Theorem 11 ($t = 3$ rigidity)**.** *If $t = 3$ and both cells are nonempty, then $n_A + n_B \leq 57$, and $(1, 56)$ is attained. By contrast $t = 1$ and $t = 2$ have no such penalty: both-nonempty pairs attain 64.*

(PROVEN, with the folded-7-cube connectivity input MACHINE-verified; the exact both-nonempty maxima 57/64/64 *are also directly MACHINE-computed (`stage3.py`, `stage6.py`))*

Proof. The folded 7-cube is vertex-transitive and 7-connected: connectivity was verified exactly by unit-capacity max-flow (Menger) over all non-neighbor pairs from a base vertex, which suffices by transitivity (`stage3.py`); it is also an instance of the theorem that the vertex connectivity of a distance-regular graph equals its valency [5]. Now let $S_A, S_B \neq \emptyset$ be the two sides of an independent set. Since $S_B \neq \emptyset$, $N[S_A] \neq V$; then the boundary $\partial S_A = N[S_A] \setminus S_A$ separates S_A from $V \setminus N[S_A]$, so $|\partial S_A| \geq 7$ by 7-connectivity. Hence $n_A + n_B \leq |S_A| + \big(64 - |S_A| - 7\big) = 57$. For tightness take S_A a single vertex ($|\partial S_A| = 7$ exactly) and S_B the complement of its closed ball: $1 + 56 = 57$. For $t = 1$: one color class of B plus the other fifteen classes of A gives $4 + 60 = 64$ with both cells nonempty; similarly for $t = 2$ $(2 + 62)$. □

The moral of Theorem 11 is a warning: the "odd overlap" regime is not uniformly expensive. Odd $t = 3$ carries a real rigidity penalty; odd $t = 1$ is combinatorially as cheap as the even case.

7.2 Localization caps

Lemma 12 ($W_{\leq 6}$ localization)**.** *Any legal configuration whose occupied cells all lie inside a fixed point set W with $|W| \leq 6$ has at most* 64 *lines. (PROVEN)*

Proof. Let $D = [12] \setminus W$, $|D| = 12 - |W| \geq 6$. Every occupied cell $A \subseteq W$ has support $H \supseteq D$. Classify lines by the restriction of their sign vector to D modulo global sign: $2^{|D|-1}$ classes, and (as in Theorem 9) each cell holds exactly $2^{8-|D|-1}$ lines per class. Two same-class lines in *distinct* cells $A, B \subseteq W$ conflict outright: their common support $R \supseteq D$ has $|R \setminus D| = |W| - |A \cup B| \leq 6 - 5 = 1$, and the restriction sum over D is $\pm|D|$, so $\left|\sum_R \varepsilon\varepsilon'\right| \geq |D| - 1 \geq 5$. Hence each class contributes lines of at most one cell, i.e. at most $2^{8-|D|-1}$ lines, and the total is at most $2^{|D|-1} \cdot 2^{8-|D|-1} = 64$. One full cell attains it. □

Lemma 13 (W_7 localization)**.** *Any legal configuration whose occupied cells lie inside a fixed* 7*-point set has at most* 64 *lines, attained. (MACHINE: with $|D| = 5$ the same-class argument no longer gives outright conflict, but dropping cross-class edges only relaxes the problem; the* 16 *fold classes on D (each an exact* 140*-vertex sub-problem over the* 35 *cells) have exact maximum independent sets of size* 4 *each, summing to* 64 *(`stage7.py`, `w7_result.json`; about two seconds))*

Lemma 14 (W_8 localization)**.** *Any legal configuration whose occupied cells lie inside a fixed* 8*-point set W has at most* 128 *lines, attained. (PROVEN from Lemma 13)*

Proof. W has eight 7-point subsets W'; a cell $A \subseteq W$ ($|A| = 4$) lies in exactly 4 of them (choose which of the 4 points of $W \setminus A$ to delete). Summing the Lemma 13 bound over the eight W' counts every line exactly 4 times: $4N \leq 8 \cdot 64$, i.e. $N \leq 128$. Attained by a complementary pair of cells A and $W \setminus A$ ($t = 0$, conflict-free): $64 + 64$. □

Corollary 15 (full-cell collapse)**.** *If some cell is full ($n_A = 64$) then every cell meeting A is empty, all other occupied cells lie inside the* 8*-set H_A, and the configuration has at most $64 + 128 = 192$ lines. Consequently every configuration with ≥ 193 lines — in particular any hypothetical* 289 *— has $n_A \leq 63$ in every cell. (PROVEN)*

Proof. By Corollary 10, $n_A = 64$ forces $n_B = 0$ for every B with $B \cap A \neq \emptyset$; the surviving cells satisfy $B \cap A = \emptyset$, i.e. $B \subseteq H_A$, and Lemma 14 caps them at 128. □

7.3 The even side: classification, closure, and the equivalence

Call a family of cells *even* if all pairwise intersections $|A \cap B|$ are even (i.e. $t \in \{0, 2\}$ — support overlaps $|H \cap H'| \in \{4, 6\}$). The slice-aligned cell families of Theorem 6 are even, but the converse fails, and this failure is precisely the hole in our earlier roadmap. The classification is, pleasingly, the $d/e_7/e_8$ trichotomy familiar from doubly-even binary codes and root lattices.

Theorem 16 (even classification)**.** *Let $\mathcal{F}$ be an even family of cells, and make $\mathcal{F}$ a graph by joining $A \sim B$ when $|A \cap B| = 2$. Then the members of each connected component $\mathcal{F}_0$, with point support $P = \bigcup \mathcal{F}_0$, form one of:*

- (d) *a pair-aligned family: P partitions into 2-point blocks such that every member of $\mathcal{F}_0$ is a union of two blocks;*
- (e7) *a subfamily of the 7 complements of the lines of a Fano plane on P, $|P| = 7$;*
- (e8) *a subfamily of the 14 weight-4 codewords of an extended-Hamming-type $[8, 4, 4]$ code on P, $|P| = 8$.*

Moreover the supports of distinct components are disjoint. (PROVEN)

Proof. Disjointness of component supports: if $x \in P_1 \cap P_2$ then $x \in A \cap B$ for some $A \in \mathcal{F}_1$, $B \in \mathcal{F}_2$, forcing $|A \cap B| = 2$ and an edge across components.

Fix a component $\mathcal{F}_0$. Say a member C *splits* the block $p = A \cap B$ (where $|p| = 2$) if $|C \cap p| = 1$.

Case 1: no member splits any block. We claim the pairwise intersections form a disjoint block family. Suppose $p = A \cap B$ and $q = C \cap D$ with $|p \cap q| = 1$, say $p = \{x, y\}$, $q = \{x, z\}$, $y \neq z$. Since $C \supseteq q \ni x$ and C does not split p, also $y \in C$; likewise $y \in D$; but then $\{x, y, z\} \subseteq C \cap D = q$, impossible. So distinct blocks are disjoint. Each member A meets other members in blocks contained in A, pairwise disjoint, hence in 0, 1 or 2 whole blocks; the remaining $4 - 2(\#\text{blocks in } A)$ points of A lie in no other member and can be grouped into fresh blocks arbitrarily. The result is a pairing of P with every member a union of two blocks: type (d).

Case 2: some C splits $p = A \cap B = \{x, y\}$, with $x \in C$, $y \notin C$. Evenness forces $|C \cap A| = 2$, say $C \cap A = \{x, a\}$ with $a \in A \setminus p$, and $|C \cap B| = \{x, b\}$ with $b \in B \setminus p$; the fourth point c of C lies outside $A \cup B$. Writing $A = \{x, y, a, a'\}$, $B = \{x, y, b, b'\}$, $C = \{x, a, b, c\}$, the triple spans exactly 7 points $S = \{x, y, a, a', b, b', c\}$. Over $\mathbb{F}_2$, the span $V_3 = \langle A, B, C\rangle$ has dimension 3 and its 7 nonzero words

$$A,\ B,\ C,\ A{+}B = \{a, a', b, b'\},\ A{+}C = \{y, a', b, c\},\ B{+}C = \{y, b', a, c\},\ A{+}B{+}C = \{y, a', b', c\}$$

all have weight 4. A $[7, 3]$ code with all nonzero weights 4 is the Fano-complement code: complements of the words are 3-sets that pairwise meet in exactly $|v \cap w| - 1 = 1$ point (evenness forces $|v \cap w| = 2$), and $7 \cdot \binom{3}{2} = 21 = \binom{7}{2}$ pairs means every point pair lies on exactly one 3-set — a Steiner $S(2, 3, 7)$, i.e. a Fano plane on S, its lines the complements. (Such classifications of codes generated by weight-4 words go back to [15].)

Now take any further member D of $\mathcal{F}_0$. Evenness against A, B, C makes $\mathbf{1}_D$ orthogonal to V_3, so $D \cap S \in V_3^{\perp}\restriction_S = V_3 \oplus \langle \mathbf{1}_S\rangle$, whose weights are $\{0, 4\} \cup \{3, 7\}$; since $|D| = 4$, either $D \subseteq S$ and $D \in V_3$ (a Fano complement), or $|D \cap S| = 3$ with $D \cap S$ a Fano line L and $D = L \cup \{z\}$, $z \notin S$, or $D \cap S = \emptyset$. A member disjoint from S has intersection of size ≤ 1 with every member meeting S, so no edge connects it to them, and connectivity confines $\mathcal{F}_0$ to members meeting S.

If no member takes the $L \cup \{z\}$ form, all members lie in S: type (e7). Otherwise fix $D = L \cup \{z\}$ and set $S' = S \cup \{z\}$, $V_4 = \langle V_3, D\rangle$. For a Fano complement $v = S \setminus L'$: $|v \cap D| = |v \cap L| = 3 - |L \cap L'|$, which is 2 for $L' \neq L$ (weight of $v+D$ is 4) and 0 for $L' = L$ (then $v + D = \mathbf{1}_{S'}$). So V_4 is a $[8,4]$ code on S' containing $\mathbf{1}_{S'}$ with all other nonzero words of weight 4 — an extended-Hamming-type $[8,4,4]$ code, self-dual and doubly even, with exactly 14 weight-4 words. Any further member E is orthogonal to V_4, so $E \cap S' \in V_4^{\perp} = V_4$ with $|E \cap S'| \in \{0,4\}$: either E is one of the 14 codewords ($E \subseteq S'$), or $E \cap S' = \emptyset$ — and in the latter case E again meets every S'-member in $\leq 1 < 2$ points, so connectivity excludes it. Note also that a second splitting member $D' = L' \cup \{z'\}$ with $z' \neq z$ is impossible at this stage, since $|D' \cap S'| = 3 \notin \{0,4\}$. Hence all members lie in S': type (e8). □

Theorem 17 (even closure)**.** *Every legal configuration whose occupied cells form an even family has at most* 288 *lines, and any such configuration with* 288 *lines is slice-aligned: its occupied cells are P-aligned for some pairing P. The branch caps are: pair-aligned* ≤ 288*; a component of type (e7) confines the whole configuration to* ≤ 128 *lines; type (e8) to* ≤ 192*. (PROVEN, given the solver-certified (CP-SAT OPTIMAL) slice cap of Theorem 6(b) and the MACHINE caps of Lemmas 13–14)*

Proof. Decompose the occupied cells into components as in Theorem 16; supports are disjoint, and cells of distinct components are conflict-free ($t = 0$), so the configuration is a free union over components.

If some component is of type (e7) with support S (7 points): its lines are capped by Lemma 13 at 64. All other occupied cells lie in the remaining 5 points, which host at most one cell (two distinct 4-subsets of a 5-set meet in 3 points — odd — contradicting evenness), for at most 64 more: total ≤ 128.

If some component is of type (e8) with support S' (8 points): Lemma 14 caps it at 128; the remaining 4 points host at most the single cell $[12] \setminus S'$: total ≤ 192. (Two components of types e7/e8 cannot coexist: $7+7, 7+8, 8+8 > 12$.)

Otherwise every component is pair-aligned. Their pairings live on disjoint supports; extend the union of the pairings arbitrarily to a full pairing P of $[12]$ (the leftover point count is even). Every occupied cell is then P-aligned, and Theorem 6(b) caps the configuration at 288. Since $128, 192 < 288$, a 288-line even configuration must be of this slice-aligned kind. □

Corollary 18 (the odd pair is unavoidable)**.** *Every legal half-integer configuration with at least* 289 *lines contains two occupied cells with odd overlap ($t \in \{1,3\}$, support overlap $|H \cap H'| \in \{5,7\}$). (PROVEN, given the CP-SAT-OPTIMAL slice cap of Theorem 6(b))*

Corollary 19 (equivalence)**.** *The odd-overlap lemma ("every legal configuration containing an occupied odd-overlap cell pair has at most* 288 *lines") is equivalent to $\alpha(U) = 288$. (PROVEN, given the CP-SAT-OPTIMAL slice cap of Theorem 6(b))*

Proof. If the lemma holds, take any configuration with ≥ 289 lines: by the lemma its occupied cells form an even family, and Theorem 17 caps it at 288 — contradiction; with the 288 witness, $\alpha(U) = 288$. The converse is trivial. □

Remark 20 (a repaired hole in the roadmap). Our campaign roadmap (artifact `K2_novelty.json`, consequences field — a campaign artifact, not shipped with this note and distinct from the shipped `certs/novelty.json`) asserted at cell level that a configuration escaping every slice must contain an odd-overlap cell pair. That assertion is *false*: the e7 and e8 families of Theorem 16 are even families aligned with no pairing — cell-level counterexamples. The even-closure theorem repairs the roadmap: what is true is not that even families are slice-aligned, but that the non-slice-aligned even families (e7/e8) are *small* — capped at 128/192 lines, far below 288. An earlier draft of this note flagged the slice-like claim as an unproven heuristic; Theorem 17 replaces it.

7.4 Cell triples: Venn structure, not overlap type, sets the cap

The pair cap (64) and the localization caps invite a systematic triple study. Cell triples fall into exactly 25 orbits under S_{12}, and the canonicalized Venn signature $\big(|A\cap B\cap C|, |A\cap B\setminus C|, |A\cap C\setminus B|, |B\cap C\setminus A|\big)$ is a complete orbit invariant. (MACHINE: exhaustive orbit enumeration, `stage4b.py`) The exact maximum independent set of each 192-line triple world:

triple class	orbits	exact cap	status
all three pairwise disjoint	1	192	trivial (PROVEN)
some disjoint pair, not all	7	128	(PROVEN), Prop. 21
all overlapping, $\lvert A\cup B\cup C\rvert \le 7$	10	64	localization (Lemmas 12–13)
all overlapping, $\lvert A\cup B\cup C\rvert \ge 8$, non-sunflower	4	64	(MACHINE), exact B&B
point-core sunflower, t's $(1,1,1)$, union 10	1	80	(MACHINE), exact
pair-core sunflower, t's $(2,2,2)$, union 8	1	80	(MACHINE), exact
Venn $(1,0,0,1)$, t's $(1,1,2)$, union 9	1	$[78,80]$	(OPEN) interval

Proposition 21. *If a triple $\{A,B,C\}$ contains a disjoint pair, say $A\cap B=\emptyset$, but is not totally disjoint, then $n_A+n_B+n_C\le 128$, attained ($n_C=0$, A,B full). (PROVEN)*

Proof. If C meets both: Corollary 10 on both overlapping pairs gives $(n_A+n_C)+(n_B+n_C)\le 128$, so $n_A+n_B+n_C\le 128-n_C\le 128$. If C meets only A: $n_B\le 64$ and $n_A+n_C\le 64$. □

The MACHINE rows are exact branch-and-bound with verified independence of the optimal witnesses; the 80-caps were re-verified for this note in under a second each. The one OPEN entry has a verified 78-line witness and a δ-class upper bound of 80. The structural moral is recorded because it killed a natural proof strategy: *the multiset of pairwise overlap types does not determine the cap* — $(1,1,1)$ occurs with cap 64 (Venn $(0,1,1,1)$) and with cap 80 (the point-core sunflower). Three of the 25 classes beat 64: the two sunflowers with disjoint petals (both exactly 80) and the OPEN Venn $(1,0,0,1)$ class, whose verified 78-line witness already exceeds 64; any cut library built on pairs and triples must carry the 80's.

7.5 The LP barrier: why cut aggregation cannot finish

The ladder (Section 6), the pair caps, the localization caps, the triple caps, and the slice cuts are all *valid occupancy inequalities*: linear inequalities $\sum_A c_A n_A\le b$ satisfied by the occupancy vector of every legal configuration. It is natural to hope that some finite library of such certified facts, fed to an LP, proves $\alpha(U)\le 288$. That hope is provably dead:

Theorem 22 (LP barrier)**.** *Let $\mathcal{I}$ be any family of valid occupancy inequalities in which each member has at most 109 cells with positive coefficient. Then the linear program $\max\sum_A n_A$ subject to $\mathcal{I}$ and the box constraints $0\le n_A\le 64$ has optimum at least 31,680/109 >290.6. In particular no such family certifies $\alpha(U)\le 289$, let alone 288 — where a family certifies $\alpha(U)\le b$ if its LP optimum is below $b+1$, so that flooring the optimum proves the integer bound (the convention of the ϑ ceilings of Section 8); an optimum above 290.6 leaves every $b\le 289$ unreachable. (PROVEN)*

Proof. Consider the uniform fractional point $x_A \equiv 64/109$, which satisfies the box constraints. Let $\sum_A c_A n_A \le b$ be a valid inequality whose positive support $\mathcal{S} = \{A : c_A > 0\}$ has $|\mathcal{S}| \le 109$ (if $\mathcal{S} = \emptyset$, validity on the empty configuration gives $b \ge 0$ and the point trivially satisfies it). For each $A \in \mathcal{S}$ the single-cell configuration "cell A full" is legal, so $64\, c_A \le b$; averaging over $\mathcal{S}$,

$$\sum_A c_A x_A \;\le\; \sum_{A\in\mathcal{S}} c_A \frac{64}{109} \;\le\; \sum_{A\in\mathcal{S}} c_A \frac{64}{|\mathcal{S}|} \;=\; \frac{1}{|\mathcal{S}|}\sum_{A\in\mathcal{S}} 64\, c_A \;\le\; b.$$

So x is feasible, with objective $495 \cdot 64/109 = 31{,}680/109 \approx 290.64$. $\square$

Corollary 23. *Every cut family in this note has positive support far below the threshold: pair cuts (2 cells), triple cuts (3), $W_{\le 6}/W_7$ localization (≤ 35), e7/e8 cuts (7/14), slice cuts (15), ladder cuts through level 3 (≤ 18). An LP over all cuts of support ≤ 35 has optimum $\ge 31{,}680/35 > 905$; adding the W_8 cuts (support 70) still leaves ≥ 452. Any LP/cutting-plane proof of $\alpha(U) \le 288$ must use valid inequalities whose positive support covers at least 110 of the 495 cells — and since $31{,}680/110 = 288$ exactly, support-110 families are the theoretical minimum.* (PROVEN)

Two honest remarks. First, the barrier binds the *LP aggregation* paradigm: a combinatorial case analysis may of course still use small-support facts (the ladder's own Theorem 8 is such a statement, and complete orbit enumeration to the diameter would still constitute a proof — the barrier says the enumeration cannot be short-circuited by an LP). Second, the barrier explains Table 2's third row structurally: the 10,395 slice cuts have support 15 each, so no reweighting of them could ever certify anything below $31{,}680/15 = 2112$; their LP value 9504 is not a solver failure but a theorem-level impossibility.

7.6 The star reduction: a global inequality that passes the barrier

The barrier demands valid inequalities with large support. There is a natural family: one inequality per *coordinate*.

For $x \in [12]$ let the *star* of x be the set of lines whose cell (defect 4-set) contains x: $\mathrm{star}_x = \{(A, u) : x \in A\}$, comprising the $\binom{11}{3} = 165$ cells through x and $165 \cdot 64 = 10{,}560$ lines. Define the U_{11} *world* as the half-integer world on ground set [11]: cells the 165 three-subsets A' of [11] (defects of 8-point supports), 64 even-sign lines per cell, conflict iff $\left|\sum_{i\in H\cap H'} \varepsilon_i \varepsilon'_i\right| \ge 5$.

Theorem 24 (star reduction). (i) *For every x, the induced conflict graph on star_x is isomorphic to the U_{11} world via $A \mapsto A \setminus \{x\}$: the conflict predicate never reads the coordinate x. In U_{11} every cell pair has common support $r = 5 + |A' \cap B'| \in \{5, 6, 7\}$ — there are no free cell pairs, so the pair cap $n_{A'} + n_{B'} \le 64$ binds on all $\binom{165}{2}$ pairs, and the entire cut library of this section transfers verbatim.* (PROVEN)

(ii) *Every line lies in exactly 4 stars, so for any legal configuration S, $4|S| = \sum_{x\in[12]} |S \cap \mathrm{star}_x| \le 12\, \alpha(U_{11})$, i.e.*

$$\alpha(U_{12}) \;\le\; 3\, \alpha(U_{11}).$$

(PROVEN)

(iii) *The 288-line witness of Theorem 1 meets every star in exactly 96 lines.* (MACHINE: *recomputed from the shipped witness certificate by* `make verify-structure`; *uniform* 96×12) *Consequently $\alpha(U_{11}) \ge 96$, and the reduction (ii) is exactly tight on the witness: $4 \cdot 288 = 12 \cdot 96$.*

(iv) *If* $\alpha(U_{11}) \le 96$ *then* $\alpha(U_{12}) = 288$*: the odd-overlap lemma holds, the ladder's gap closes, and the* $W_2 \cup U$ *architecture is globally sealed at* 420 *lines. (PROVEN, given (ii) and the witness)*

(v) *Conversely, by pigeonhole any* 289*-line configuration meets some star in at least* $\lceil 4{\cdot}289/12 \rceil =$ 97 *lines: every record attempt in the half-integer world contains a* 97*-line* U_{11} *configuration. (PROVEN)*

Proof. (i) A cell $A \ni x$ has support $H = [12] \setminus A \subseteq [12] \setminus \{x\}$; identifying $[12] \setminus \{x\}$ with $[11]$, cells through x correspond bijectively to 3-subsets $A' = A \setminus \{x\}$, supports and sign classes are preserved, and the conflict predicate depends only on $H \cap H' \subseteq [11]$ and the sign vectors. For the overlap census: $|A' \cap B'| \in \{0, 1, 2\}$ gives $r = 11 - |A' \cup B'| = 5 + |A' \cap B'| \ge 5$. (ii) $|A| = 4$: a line's cell contains exactly 4 coordinates, so the line lies in exactly 4 stars; each $S \cap \mathrm{star}_x$ is independent in the star_x world, hence of size $\le \alpha(U_{11})$ by (i). (iii) is a finite recomputation from `certs/config420.json`. (iv) $288 \le \alpha(U_{12}) \le 3 \cdot 96 = 288$. (v) If all twelve stars held ≤ 96, the sum $4 \cdot 289 = 1156 > 12 \cdot 96 = 1152$ fails. □

Remark 25 (the star cuts pass the barrier). Each star cut $\sum_{A \ni x} n_A \le \alpha(U_{11})$ has positive support $165 \ge 110$ cells — the first valid family we possess on the far side of Theorem 22 — and if the cap 96 holds, the twelve star cuts alone close the world: $4 \sum_A n_A = \sum_x \sum_{A \ni x} n_A \le 12 \cdot 96$ gives exactly 288. The uniform witness saturation (iii) shows they are tight, so no slack is wasted: the star family is not merely sufficient, it is exactly calibrated.

The section's conclusion is a genuine reduction of both open questions of this note to one:

Problem 26 (the U_{11} problem)**.** *Is* $\alpha(U_{11}) \le 96$*? Equivalently (given the witness): does the* 10,560*-line,* 165*-cell* U_{11} *world — three times smaller than* U_{12}*, with no free cell pairs — have independence number exactly* 96*? A proof closes* $\alpha(U_{12}) = 288$ *and hence the odd-overlap lemma and the* 420*-line global cap (Theorem 24(iv)); a* 97*-line* U_{11} *witness would refute the star route and is a necessary ingredient of any* 842*-point record in this architecture (Theorem 24(v)), though not by itself sufficient. (OPEN)*

8 The architecture sweep: every enumerable frame family closes low

The slice-closure results seal one architecture. In parallel we ran a closing sweep over every other frame architecture we could enumerate, for both record targets (303 lines = 606 points in $\mathbb{R}^{11}$, beating 604; 421 lines = 842 points in $\mathbb{R}^{12}$, beating 841). Each candidate family is compiled to an exact conflict graph over its ring ($\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{5})$; the adjudicator works over $\mathbb{Q}(\sqrt{d})$ for square-free d and passes an adversarial self-test), then closed by one of: exact ILP duality (HiGHS [10], integer optimal), or a *certified* Lovász ϑ bound [13]. The ϑ certification works as follows (`kissing_frame_adjudicate.py`, a campaign tool that is not part of this repository): the dual $\vartheta(G) = \min_Y \lambda_{\max}(J + Y)$ (Y symmetric, supported on edges) is solved numerically; Y is then rounded to dyadic rationals (denominator 2^{16}) so that $B = J + Y$ is *exactly* representable in binary floating point, and a dyadic $t_{\mathrm{cert}} \ge \lambda_{\max}(B)$ is certified by a shifted floating-point Cholesky factorization of $t_{\mathrm{cert}} I - B$: the factorization succeeds with a shift of 2^{-16}, which exceeds 100 times the Demmel–Rump rounding radius $\gamma \|M\|_\infty$, $\gamma = (n{+}1)u/(1 - (n{+}1)u)$, $u = 2^{-53}$, so $\lambda_{\min}(t_{\mathrm{cert}} I - B) > 0$ rigorously (up to IEEE-754 conformance of the hardware). Since any such B

cand.	d	family	ring	$\lvert V\rvert$	$\lvert E\rvert$	lines: found / ceiling	verdict
B2r	11	D_4-root frame $\mathbb{R}^7 \oplus \mathbb{R}^4$, 45°-rotated	$\mathbb{Q}(\sqrt{2})$	803	13,860	87 / $\leq$ 94 (ϑ)	closed
B3	11	icosahedral 3-frame, pure-$\sqrt{5}$ sector	$\mathbb{Q}(\sqrt{5})$	574	8,960	126 / = 126 (ILP)	closed
B5	11	Steiner $S(4,5,11)$ weight-5 world, M_{11}	$\mathbb{Q}(\sqrt{5})$	1067	66,000	66 / $\leq$ 77 (ϑ)	closed
B7	11	D_5-root frame $\mathbb{R}^6 \oplus \mathbb{R}^5$, 45°-rotated	$\mathbb{Q}(\sqrt{2})$	746	11,060	70 / = 70 (ILP)	closed
K5	12	8+4 split modular, joint glue$\leftrightarrow$side	$\mathbb{Q}(\sqrt{2})$	1988	—	324 / = 324 (ILP)	closed

Table 3: The Round-88 closing sweep (artifact `SWEEP_VERDICTS.json`). Targets: 303 lines ($d = 11$), 421 lines ($d = 12$). "=" entries are exact integer optima (HiGHS primal = dual); "$\leq$" entries are certified ϑ ceilings with the best independent set found alongside. Numeric ϑ values before flooring: 94.14 (B2r), 77.0 (B5), 80.76 (B7, superseded by the exact ILP 70). K5's pool is an induced subgraph of the grand mixed graph of Section 2, so its optimum is also $\leq$ the 420 of that world.

is a feasible ϑ dual matrix, $\alpha(G) \leq \vartheta(G) \leq t_{\mathrm{cert}}$, and the stated integer ceiling $\lfloor t_{\mathrm{cert}} \rfloor$ is a theorem for the graph, not a solver report.

Together with the earlier waves — Bianchi pair-system neighborhood (d11) closed rigorously at 302 over all 30 enumerated pair systems; Hadamard weight-8 companion (d11) at 267; the Bianchi-port (d12) at 359; the unrotated D_4 frame (d11) at 299; and the grand mixed world (d12) at 420 (this note) — every frame architecture we can currently write down is closed far below its target. Two rigorous $d = 11$ anchors frame the picture: the pure whole-cell world caps at $11 + 8 \cdot D(11,4,3) = 11 + 8 \cdot 35 = 291$ (with $D(11,4,3) = 35$ exact by ILP), and the Bianchi-type extremum over all enumerated pair-system frames is 302 — precisely the published record's architecture, sitting at the top of its own family.

The anti-correlation. The sweep's structural lesson is that the frame-richness lever points the wrong way. A richer frame (a larger line system installed on a coordinate block, e.g. a rotated root system) gains its own lines but forces irrationality or support collisions onto the ambient coordinates, thinning the weight-4 base world that supplies the bulk of any large configuration: the rotated D_4 frame family drops from 299 (unrotated) to $\leq$ 94 (rotated); the steepest frame gain in the sweep (B7, net +15 frame lines) leaves the thinnest base (15 cells) and closes at 70. Across all candidates the total is dominated by the surviving weight-4/half-integer base, never by the frame — *frame richness anti-correlates with total capacity.* Base-preserving mixing (the $W_2 \cup U$ architecture of this note, which ties 420) is the only family that even reached the known record tier.

9 Discussion and open problems

What is new. A second, provably non-isometric 420-line antipodal configuration in $\mathbb{R}^{12}$; the first (to our knowledge) built on partial cells, found by unbiased search over a mixed-shell world and then fully explained by an exact rotation certificate; slice-closure theorems that cap its entire architecture at the record tie; a machine-checked exclusion ladder localizing where any improvement must live; an exact structure theory for the half-integer world (pair structure lemma, even closure, the equivalence of Corollary 19); an LP barrier theorem explaining why the cut paradigm cannot finish; a star reduction that collapses the frontier to the U_{11} world; and a certified sweep that seals the enumerable alternatives.

What is honestly open.

1. **The U_{11} problem (headline): is $\alpha(U_{11}) \le 96$?** (Problem 26.) By Theorem 24 this single question now carries both halves of the frontier: a proof gives $\alpha(U_{12}) = 288$ — the odd-overlap lemma (Corollary 19), the global 420-line cap for the $W_2 \cup U$ architecture, and the retirement of the ladder's gap — while a 97-line U_{11} witness is a necessary ingredient of any 842-point record in this architecture. The U_{11} world is three times smaller than U_{12} (10,560 lines, 165 cells), has no free cell pairs, and inherits the entire certified cut library of Section 7 ($\kappa = 64$ on every pair, t=3 rigidity 57, localization 64/64/128, sunflower 80's, e7/e8 caps 64/128). The natural attack is two-sided: a KaMIS lower hunt for 97, and a CP-SAT ladder on U_{11} seeded with those cuts — mindful that Theorem 22 (which applies verbatim to U_{11}: certifying $\alpha(U_{11}) \le 96$ requires positive support at least $\lceil 10{,}560/97 \rceil = 109$ of the 165 cells) again forbids a small-support LP finish there.
2. **The odd-overlap lemma, now an equivalence.** Prove: every legal half-integer configuration containing a cell pair with $|H \cap H'| \in \{5, 7\}$ has at most 288 lines. Section 7 proved this is not merely sufficient but *equivalent* to $\alpha(U) = 288$ (Corollary 19), and that any 289-line configuration must contain an occupied odd pair (Corollary 18). The barrier theorem constrains the proof shape: no aggregation of the small-support facts we hold can do it; the star route (Problem 26) is the one global inequality family known to be both valid and tight.
3. **Orbit enumeration at levels 4 and 5.** Quadruple orbits are within reach (18–21-cell universes solve to OPTIMAL in minutes), and each further level shrinks the search region for a 289; complete enumeration to the diameter would still constitute a proof. Theorem 22 tempers the ambition: bounded-level ladder certificates are support-$\le$ $(15{+}k)$ cuts, so no LP over them closes the world — the enumeration must actually reach exhaustion, or be replaced by the U_{11} route.
4. **Global $\alpha(U)$ relaxations.** Is there any tractable relaxation adapted to the 64-per-cell-pair conflict structure that beats the gaps of Table 2? The barrier theorem sharpens the question: any useful valid-inequality family must have $\ge$ 110-cell positive support. The twelve star cuts are the first such family; finding a *second*, independent one (weighted stars? spectral cuts on the U_{11} quotient?) would over-determine the polytope near the witness and might close the gap without solving Problem 26 exactly.
5. **The 842 frontier.** An antipodal 842-point configuration means 421 lines at coherence $\frac{1}{2}$; the LP ceiling is 614 lines [17], so geometry leaves 46% headroom, while every architecture in this note and its predecessors closes at $\le$ 420. The enumerable-frame space is exhausted, not the problem: the biquadratic $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ hybrid sector (icosahedral frames mixed with root alphabets) is untested for lack of exact tooling, and a closed-form negative theorem for the weight-4-based design space ("no such architecture reaches the target") is supported by every data point but unproven. Within the half-integer route specifically, the record hunt is now the 97-in-U_{11} hunt of Problem 26.
6. **Classification at 420.** Are the classical configuration and the configuration of Theorem 1 the only 420-line systems in $\mathbb{R}^{12}$ up to isometry? The rotation certificate shows both live in the integer world up to isometry; a classification of maximum antipodal configurations there (whole and partial cells) may be within reach of the counting method, now supplemented by the even-classification trichotomy of Theorem 16.

Relation to the record. Nothing in this note improves $k(12) \geq 841$ [18]; the new configuration ties the *antipodal* record of 840 points with a second, inequivalent object, and the closure theorems say that this entire corner of the landscape is now understood exactly: 421 needs either odd-overlap mixing in the half-integer world — provably unavoidable (Corollary 18), against the ladder's evidence, and now concentrated in the single question of Problem 26 — or an architecture nobody has written down yet.

A Reproduction and verification

The repository accompanying this note contains

`certs/config420.json`	the 420-line witness (line descriptions only)
`certs/rotation.json`	the pairing of Theorem 5, nothing else
`certs/novelty.json`	the family parameters of Theorem 2, nothing else
`certs/receipts/`	solver receipts: CP-SAT ladder levels 0–3 (level 1 inside the v2 decision record), search beyond level 3, the 288 warm start, the v2 decision record, the edge census, and the Table 3 verdicts (`SWEEP_VERDICTS.json`)
`scripts/verify_420.py`	packing + counts checker (stdlib; no floats)
`scripts/verify_rotation.py`	$\mathbb{Q}(\sqrt{2})$ isometry + $A \cup W_4$ membership
`scripts/verify_novelty.py`	counting-theorem checker
`scripts/verify_structure.py`	Section 7 MACHINE-facts checker
`scripts/selftest.py`	adversarial test: corruptions must fail
`scripts/oddlemma/`	the lemma-hunt exact toolkit (verbatim)
`Makefile`	`verify` / `verify-structure` / `selftest` / `pdf`

`make verify` recomputes, in exact integer and $\mathbb{Z}[\sqrt{2}]$ arithmetic with nothing echoed from the certificate files: (i) all 87,990 pairwise predicates (1) for the witness, its composition, cell table and histogram (Theorem 1); (ii) the histogram formulas and the five-step contradiction of Theorem 2, with every interaction constant derived by finite computation inside the checker; (iii) the rotation image of every witness line under T_{P_0}, the 60° bound on every image pair, and the 47+8 cell census (Theorem 5). `make verify-structure` additionally recomputes every MACHINE entry of Section 7 from the conflict predicate alone (structure-lemma censuses 256/128/512, pair cap $\kappa = 64$, t=3 rigidity 57 with folded-7-cube connectivity 7, W_7 and corner localization, e7/e8 caps 64/128, the sunflower 80's, the uniform 96×12 star saturation of the witness, and the U_{11}-world census), in about five seconds of stdlib branch-and-bound. The CP-SAT OPTIMAL statuses underlying Theorems 6(b) and 8 and the certified ϑ/ILP ceilings of Table 3 are solver results: the receipts ship with the repository (`certs/receipts/`), but they are not stdlib-recomputable and are labeled accordingly wherever they are used. `make selftest` feeds the checkers corrupted variants (perturbed vectors, deleted lines, doctored histograms, a wrong pairing, sign-flipped rotation blocks) and passes only if every one is rejected with the expected diagnostic.

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